Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

Henrique José da Silva Santos

Student No.: 1180934

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Supervisor: Luís Gomes
Co-Supervisor: Rafael Silva

Evaluation Committee:
President:
[A definir]

Members:
[A definir]

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Abstract

This dissertation presents CLARION, an explainable (XAI-first) financial-alert system for retail investors in the US equity market (NYSE and NASDAQ). The system raises Telegram alerts from two independent triggers (abrupt market movements, detected as statistical anomalies, and incoming financial news) and, for every alert, exposes a fully traceable explanation: the detected event, the reasoning applied, the sources, and historical precedents. Its core is a news–market correlation engine that, given a news item, retrieves analogous historical news from a knowledge base and measures the impact those precedents had on prices, in the manner of an event study and without lookahead. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components (sentence-embedding retrieval, statistical anomaly detection and transparent explanation) in a functional, explainable and reproducible system. On real data, semantic retrieval clearly outperformed lexical and trivial baselines (cross-ticker precision@5 of 0.51, versus 0.35 for a lexical baseline and 0.24 for random retrieval), and the volatility-normalised detector fired far more consistently across stocks than a fixed threshold. The results are honest and reproducible, with limitations and future work stated explicitly.

Keywords: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Resumo

Esta dissertação apresenta o CLARION, um sistema de alertas financeiros explicável (XAI-first) para investidores de retalho no mercado acionista dos Estados Unidos (NYSE e NASDAQ). O sistema envia alertas via Telegram a partir de dois gatilhos independentes (movimentos abruptos de mercado, detetados como anomalias estatísticas, e novas notícias financeiras) e, para cada alerta, expõe uma explicação totalmente rastreável: o evento detetado, o raciocínio aplicado, as fontes e os precedentes históricos. O seu núcleo é um motor de correlação notícia–mercado que, dada uma notícia, recupera notícias históricas análogas de uma base de conhecimento e mede o impacto que esses precedentes tiveram nos preços, à maneira de um estudo de evento e sem lookahead. Enquanto contribuição de Engenharia de Inteligência Artificial, o trabalho integra, aplica e avalia criticamente componentes existentes (recuperação por embeddings de frases, deteção estatística de anomalias e explicação transparente) num sistema funcional, explicável e reproduzível. Em dados reais, a recuperação semântica superou claramente as baselines lexical e triviais (precision@5 cross-ticker de 0,51, face a 0,35 de uma baseline lexical e 0,24 do acaso) e o detetor normalizado pela volatilidade disparou de forma muito mais consistente entre ações do que um limiar fixo. Os resultados são honestos e reproduzíveis, com limitações e trabalho futuro explícitos.

Palavras-chave: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Acknowledgement

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r_t	log-return of an asset on day t	—
μ_t	rolling mean of returns	—
σ_t	rolling standard deviation of returns	—
z_t	z-score of the return on day t	—
k	anomaly threshold (in standard deviations)	—
w	rolling-window length	d

List of Acronyms

AI	Artificial Intelligence.
NASDAQ	National Association of Securities Dealers Automated Quotations.
NYSE	New York Stock Exchange.
RQ	Research Question.

Chapter 1

Introduction

1.1 Contextualization

Most Americans now own stock, directly or through retirement accounts and mutual funds (Gallup 2025), and they invest in the world's largest equity market, worth US$62.2 trillion at the end of 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025) (Figure 1.1). Trading happens mainly on the New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations (NASDAQ), and a growing share of it now comes from ordinary people using commission-free mobile apps, not from professional institutions.

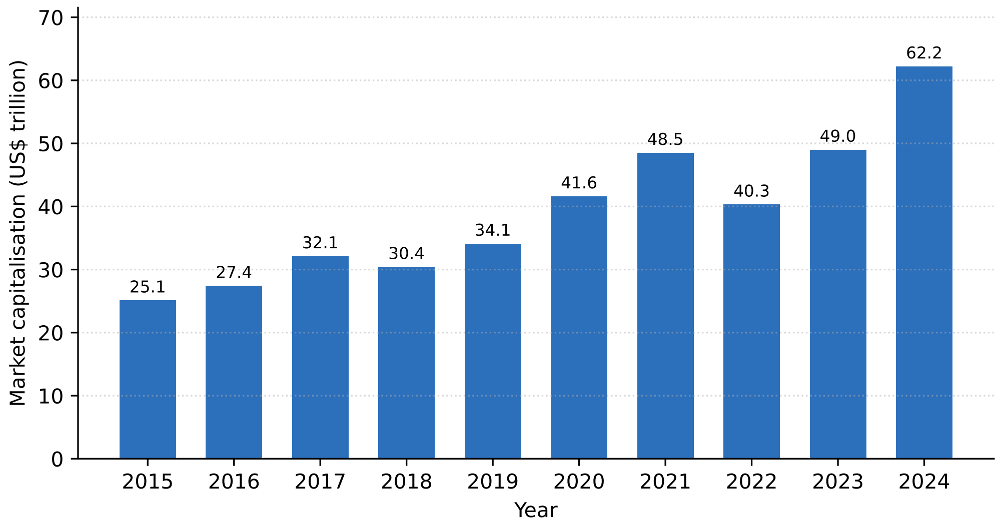

Figure 1.1: US equity market capitalisation, 2015–2024 (US$ trillion). Drawn from data reported in the SIFMA 2025 Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

These non-professional, or "retail", investors are the audience of this work, and their position is hard. Markets move continuously across thousands of securities, and relevant news arrives all through the trading day, yet retail investors have none of the tools that banks and funds take for granted. Ownership is uneven too: it reaches 87% of households earning above US$100,000 but only 28% below US$50,000 (Gallup 2025).

Meanwhile, artificial intelligence has spread quickly through finance: a 2026 survey found 81% of financial firms adopting Artificial Intelligence (AI) and 71% already using generative AI (Cambridge Centre for Alternative Finance 2026). But most of this technology is opaque, rarely showing how it reaches a conclusion (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). For someone who must act on an alert and live with the result, that opacity is the obstacle this dissertation removes: it lets the investor see *why* an alert was raised.

1.2 Problem Statement

A retail investor who wants to stay informed faces two everyday problems. The first is telling a genuinely unusual price move from ordinary volatility, which takes statistical judgement and constant attention. The second is reading what a news item might mean, which takes knowing how similar events played out before. Most people can do neither at scale, and the consumer apps that promise to help usually send bare alerts or hide their reasoning inside an opaque model.

This work takes a deliberately different stance: it does not predict prices, it *explains* events. For each abrupt move or relevant news item, the system shows what it detected, how, where the information came from, and which past events resemble the current one, together with what happened to prices afterwards (Figure 1.2).

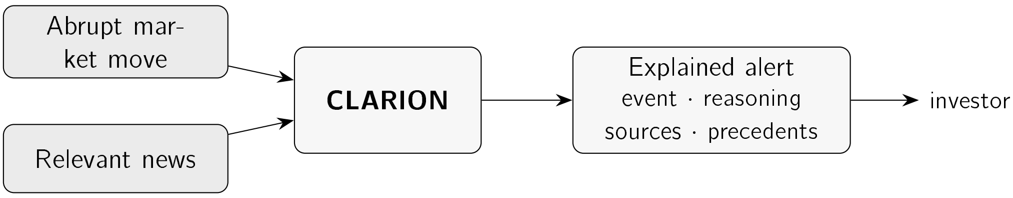

Figure 1.2: The two triggers, a sudden market move or a relevant news item, feed CLARION, which answers with an explained alert the investor can follow and check, not just a bare notification.

1.3 Research Questions and Objectives

The aim is to design, build and evaluate an explainable financial-alert system for US retail investors, driven by two independent triggers, a sudden market move and an incoming news item, and delivered over Telegram. Three research questions guide the work:

- **Research Question (RQ)1.** Can transparent statistical methods detect abrupt market movements consistently enough to be useful to a retail investor, while staying explainable?
- **RQ2.** Can a news–market correlation engine retrieve genuinely analogous historical news better than trivial baselines, and quantify the observed price impact of those precedents without lookahead?
- **RQ3.** Can the resulting explanations be made faithful to what the system actually computes, and useful to a retail investor?

These questions map onto four concrete objectives:

- build an explainable alerting pipeline driven by the two triggers;
- evaluate the anomaly detector against transparent baselines;
- evaluate the retrieval of historical news against trivial baselines;

- assess whether the explanations are faithful and useful.

The system is built incrementally, in its simplest defensible form first, a methodological choice explained in Chapter 3.

1.4 Scientific Contributions

This is a dissertation in Artificial Intelligence *Engineering*: the contribution is not a new algorithm but the disciplined assembly of existing ones into a system that works, explains itself, and can be reproduced. Three contributions stand out:

- A reproducible method for relating news to market impact, pairing sentence-embedding retrieval with event-study windows, with the similarity measure and the time horizons stated explicitly and no information about the future allowed to leak in.
- **CLARION**, an alerting system that shows its whole reasoning chain: the detected event, the supporting evidence, the sources, and the relevant historical precedents. Because each explanation is built directly from the quantities the system computes, the text cannot drift from the computation.
- A critical, honest evaluation of each core component on real data, against simple and lexical baselines, with the results, the ablations and the limitations all reported plainly.

1.5 Document Structure

The rest of the dissertation is written to be read in order, each chapter answering one question:

- **Chapter 2 (State of the Art):** what is already known, and what does this work choose?
- **Chapter 3 (Methods and Materials):** how is the system built, and how is it judged?
- **Chapter 4 (CLARION):** what is the system, and how do its data, parts, flow and decisions fit together?
- **Chapter 5 (Case Studies):** does it work on real data?
- **Chapter 6 (Conclusions):** what was learned, and what comes next?

Chapter 2

State of the Art

This chapter has one job: to justify every design choice in CLARION against the alternatives. Each section asks what a field offers and ends with what CLARION takes from it. The order is: how retail investors behave; anomaly detection; explainable AI; trust in automation; financial text representation; information retrieval; event studies; and case-based reasoning. The recurring finding is simple. The building blocks are mature, but an integrated, explainable system for the retail investor is still rare.

2.1 Retail Investing and Information Overload

How do retail investors behave, and what should an alert give them?

Behavioural finance shows that real investors depart systematically from the rational ideal (Barberis and Thaler 2003). One root cause is *loss aversion*: we feel losses more than equal gains (Kahneman and Tversky 1979), which makes investors especially sensitive to bad news, and a timely, well-explained alert valuable.

The strongest pattern is *attention*. Barber and Odean (2008) show that individuals are net buyers of attention-grabbing stocks (those in the news, with unusual volume, or extreme one-day returns): facing thousands of choices, they fall back on whatever caught their eye. Attention can even be measured through search volume (Da, Engelberg, and Gao 2011). This is decisive for the design, because a sharp price move and a salient headline are exactly CLARION's two triggers, so an alert that adds context lands where attention-driven decisions are made.

News also moves markets, and its text can be measured. Tetlock (2007) builds a daily pessimism index from a newspaper column and links it to price pressure and trading volume, an early proof that textual signals relate to market outcomes, which is the premise of CLARION's news engine. This audience is large and active too: even through the March 2020 crash, retail investors on one commission-free platform added to their holdings rather than panicking (Welch 2022), consistent with the broad ownership and market scale reported in Chapter 1 (Gallup 2025; Securities Industry and Financial Markets Association (SIFMA) 2025).

For CLARION: retail investors are attention-constrained and news-driven, so the system does not predict the market; it lowers the cost of *interpreting* the events that already grab attention, with a transparent explanation and real historical evidence.

2.2 Anomaly Detection in Financial Markets

How do you tell an abnormal price move from ordinary noise?

CLARION's first trigger is an anomaly detector, so it sits in the mature field of anomaly detection. The standard taxonomy (Chandola, Banerjee, and Kumar 2009) separates *point*, *contextual* and *collective* anomalies, and groups methods into statistical, distance/density, and machine-learning families (Figure 2.1). A sharp single-day return is a contextual point anomaly: abnormal relative to the asset's own recent behaviour.

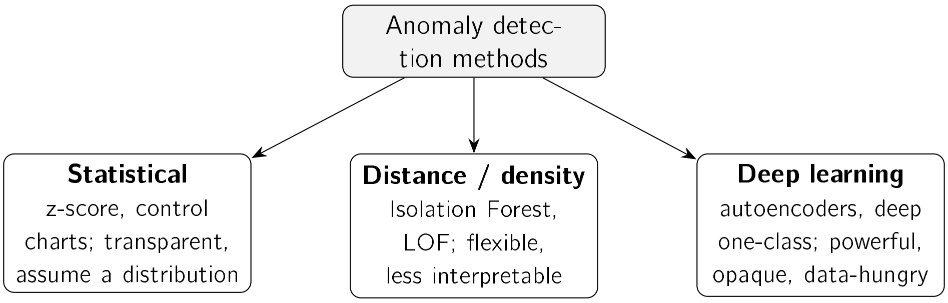

Figure 2.1: Families of anomaly-detection methods relevant to this work, following Chandola, Banerjee, and Kumar (2009) and Pang et al. (2021). Interpretability decreases, and modelling capacity increases, from left to right.

The simplest detector is a rolling z-score: flag a return that deviates far from its recent mean and standard deviation. It is attractive because it is transparent and easy to justify. Its one assumption, a locally stable distribution, is handled by the rolling window, which tracks *volatility clustering*, the well-known tendency of calm and turbulent periods to arrive in runs (Bollerslev 1986; Engle 1982). The rolling standard deviation is, in effect, a simpler non-parametric version of that idea: it adapts to the current regime without fitting a model the user could not scrutinise. A full GARCH estimate is the more sophisticated option, declined here under defensible simplicity.

More expressive detectors trade interpretability for capacity. Isolation Forest isolates outliers by random partitioning and scales to high dimensions (F. T. Liu, Ting, and Zhou 2008); the Local Outlier Factor scores how isolated a point is within its neighbourhood (Breunig et al. 2000); deep detectors learn a model of normality (Pang et al. 2021). All are powerful but harder to read off, and their edge shows most in high-dimensional data, not a single return series. In finance, detection is anyway dominated by *unsupervised* methods, because labelled anomalies are scarce (Ahmed, Mahmood, and Islam 2016), a constraint that also shapes how CLARION is evaluated (Chapter 5). Table 2.1 compares the families.

For CLARION: the signal is one low-dimensional return series that must be explained to a non-expert, so the transparent z-score wins; the more expressive families are noted as alternatives whose opacity the problem does not justify.

2.3 Explainable Artificial Intelligence

What kind of explanation should the system give?

Explainable AI (XAI) exists because the strongest models rarely show their reasoning. Surveys map the field and its central split (Adadi and Berrada 2018; Barredo Arrieta et al.

Table 2.1: Anomaly-detection approaches relevant to this work.

Approach	**How it works**	**Strengths / limitations**
Statistical z-score (rolling) (Chandola, Banerjee, and Kumar 2009)	Flags returns that deviate from a rolling mean and standard deviation	Transparent and easy to explain; assumes a locally stable distribution
Isolation Forest (F. T. Liu, Ting, and Zhou 2008)	Isolates points via random partitioning (ensemble)	Scalable, handles high dimensionality; score is not directly interpretable
Deep detectors (Pang et al. 2021)	Learn a model of normality (e.g. autoencoders)	High capacity for complex data; opaque, data-hungry, hard to justify

2020): *transparent* models, interpretable in themselves, versus *post-hoc* methods that explain a trained black box, which can be organised by what they explain (Guidotti et al. 2018) (Figure 2.2). A useful caution: "interpretability" is not one thing (Lipton 2018), so a system should say which kind it offers. CLARION offers transparency of its decision logic and traceable evidence, not a claim of full interpretability.

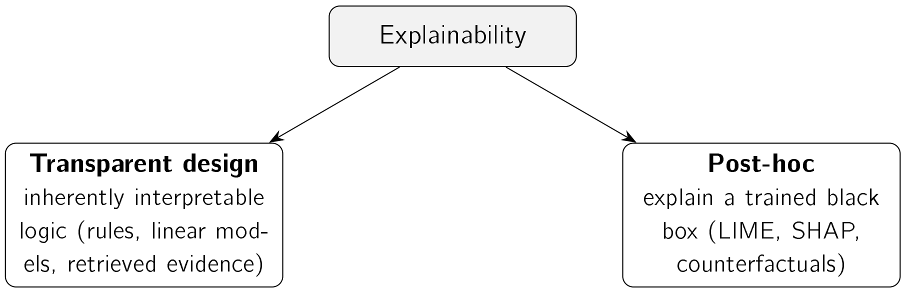

Figure 2.2: The two broad routes to explainability (Barredo Arrieta et al. 2020; Guidotti et al. 2018): building models that are transparent by design, or explaining a black box after the fact. This work follows the former.

Two post-hoc techniques are widely used: LIME fits a simple local surrogate around one prediction (Ribeiro, Singh, and Guestrin 2016), and SHAP attributes a prediction to its features using Shapley values (Lundberg and S.-I. Lee 2017). Both are valuable when one is committed to a black box. But for high-stakes decisions, Rudin (2019) argues for dropping that commitment and using models that are interpretable in the first place, since a post-hoc explanation is itself an approximation that can mislead. CLARION follows this: it explains primarily by transparent design (every quantity exposed, real precedents shown as evidence), with feature attribution as an optional extra.

Good explanations are also a human matter. People want *contrastive*, selective reasons, not an exhaustive dump (Miller 2019), which is why each alert shows a few relevant precedents and the exact quantities behind the decision. Explanations must also be evaluated, not assumed: Doshi-Velez and Kim (2017) make *fidelity*, how well an explanation reflects the real computation, a core criterion, and fidelity is the property CLARION can guarantee (Chapter 5). Table 2.2 summarises the options.

Table 2.2: Explainability approaches considered.

Method	Type	Strengths / limitations
LIME (Ribeiro, Singh, and Guestrin 2016)	Local surrogate, post-hoc	Intuitive, model-agnostic; explanations can be unstable, only locally faithful
SHAP (Lundberg and S.-I. Lee 2017)	Shapley-value attribution, post-hoc	Consistent, theoretically grounded; computationally costly
Transparent design (Barredo Arrieta et al. 2020; Rudin 2019)	Inherently interpretable logic	No approximation, faithful by construction; constrains model choice

For CLARION: pursue explainability by transparent design and judge it by fidelity; treat post-hoc attribution as a complement, not the foundation.

2.4 Trust and Appropriate Reliance in Decision Support

When should the investor actually trust an alert?

An explanation only helps if it produces *appropriate* reliance: not dismissing a sound warning, not acting blindly on a fallible one. The human-factors view (J. D. Lee and See 2004) calls the goal *calibrated* trust, reliance matched to the system's real competence, and names two failure modes, *disuse* (ignoring a good aid) and *misuse* (over-trusting a flawed one). With real money, both are costly.

Explanations are how trust gets calibrated, but they are not magic. Bansal et al. (2021) show that explanations can even cause *over-reliance*, with people accepting confident explanations of wrong answers. CLARION's response is concrete: explain with *verifiable evidence* (the actual z-score and its parameters; real precedents with their measured impact), and disclaim prediction on every alert.

For CLARION: transparency is not decoration but the mechanism for appropriate reliance, so the system shows checkable evidence rather than a persuasive story.

2.5 Financial NLP and Text Representation

How should the system represent the meaning of a headline?

To compare headlines, text must become numbers. The financial-NLP literature moves through three generations (Kearney and S. Liu 2014): lexicons, word embeddings, and contextual neural models.

Lexicons came first. Finance-specific word lists fix the fact that general sentiment dictionaries misread ordinary financial words such as "liability" or "tax" (Loughran and McDonald 2011); they are fully transparent but bounded by their vocabulary. Word embeddings then placed words in a vector space where proximity means similar use: word2vec from local context (Mikolov et al. 2013), GloVe from global co-occurrence (Pennington, Socher, and Manning 2014). Both give each word a single vector, blind to context.

The Transformer (Vaswani et al. 2017) removed that limit, and BERT (Devlin et al. 2019) produced deeply *contextual* representations. Comparing two texts with BERT directly is costly, though; Sentence-BERT (Reimers and Gurevych 2019) fixes this by giving each sentence one fixed-length vector comparable by a cheap cosine, exactly what precedent retrieval needs. At very large scale, exact search can be swapped for an approximate index (Johnson, Douze, and Jégou 2021) without changing the representation (future work). Two heavier options also exist: domain-tuned FinBERT models (Araci 2019; Y. Yang, Uy, and Huang 2020) and large financial LLMs such as BloombergGPT (Wu et al. 2023), both powerful but costlier, less transparent, and outside this work's free, reproducible constraint. Table 2.3 compares them.

Table 2.3: Text-representation approaches for financial news.

Approach	**Representation**	**Strengths / limitations**
Finance lexicons (Loughran and McDonald 2011)	Curated word lists / tone	Fully transparent; bounded by vocabulary
word2vec / GloVe (Mikolov et al. 2013; Pennington, Socher, and Manning 2014)	Static word vectors	Efficient; one vector per word, no context
BERT (Devlin et al. 2019; Vaswani et al. 2017)	Contextual token embeddings	Strong understanding; pairwise similarity is costly
Sentence-BERT (Reimers and Gurevych 2019)	Sentence embeddings (siamese)	Efficient cosine similarity; the choice in this work
FinBERT (Araci 2019; Y. Yang, Uy, and Huang 2020)	Domain-adapted language model	Finance-tuned; oriented to sentiment / communications
Financial LLMs (Wu et al. 2023)	Large generative model	Very capable; costly, opaque, outside free-tier scope

For CLARION: Sentence-BERT is the sweet spot, contextual meaning beyond lexicons and static vectors, yet cheap, comparable and free to run.

2.6 Information Retrieval and Ranking Evaluation

Once headlines are vectors, how do you rank past news, and how do you tell if the ranking is good?

Ranking a corpus by relevance to a query is the defining problem of information retrieval (IR). The vector-space model (Salton, Wong, and C. S. Yang 1975) represents documents and queries as vectors ranked by cosine similarity; CLARION's embedding retrieval is the same idea with richer vectors. Classical *lexical* ranking weights informative terms, from TF-IDF to BM25 (Robertson and Zaragoza 2009), still a strong baseline. Its known weakness is *vocabulary mismatch*: a relevant document that uses different words scores as dissimilar.

Semantic embeddings overcome exactly that, which is why CLARION compares against a lexical baseline (Chapter 5).

Because retrieval returns a ranked list, it is judged with rank-aware measures (Manning, Raghavan, and Schütze 2008); the most direct is *precision@k*, the fraction of the top k results that are relevant. CLARION adopts it because an alert can show only a handful of precedents, so what matters is whether the top few are relevant, reported against random, recency and lexical baselines.

For CLARION: retrieval reuses a mature representation-then-rank pipeline and the standard precision@k metric; the contribution is the application and the pairing with measured impact, not a new algorithm.

2.7 Event Studies and News–Market Impact

Having found analogous news, how do you measure what it did to the price, honestly?

The tool is the *event study*, introduced by Fama et al. (1969) and standardised for daily returns by Brown and Warner (1985), who set the practice of measuring returns over short post-event windows, the basis for CLARION's +1, +3 and +5-day horizons. The impact is measured strictly *after* the event, so it describes an outcome, not a forecast.

Why not just predict the price? Because public news is absorbed into prices almost at once, the efficient-market hypothesis (Fama 1970), so forecasting returns from public news is very hard by construction. CLARION therefore does not compete with market efficiency; it measures and explains the reactions that *already* followed comparable news, which is both more honest and more defensible for a non-expert.

Relating news *text* to impact is an active area: Tetlock (2007) is an early example, and a more recent strand tries to *predict* moves from news (Ding et al. 2015; Xing, Cambria, and Welsch 2018). CLARION deliberately stops short of prediction. Doing this at scale needs news aligned to prices, which is the contribution of the FNSPID dataset (Dong, Fan, and Peng 2024), used to build the knowledge base; its coverage and period limits are noted in the data card of Chapter 3.

For CLARION: pairing semantic retrieval with event-study impact over FNSPID is the methodological core, evidence about the past, never a forecast.

2.8 Case-Based Reasoning and Precedent Retrieval

What is the engine, in one familiar idea?

It is *case-based reasoning* (CBR). In the classic account (Aamodt and Plaza 1994), a CBR system *retrieves* similar past cases, *reuses* their outcomes, and optionally *revises* and *retains* them. CLARION is the retrieve-and-reuse core: each past headline plus its measured impact is a *case*; a new headline is the query; cosine similarity retrieves; the precedents' impact is reused as evidence. It stops before *revise*, so it never turns evidence into a prediction.

For CLARION: naming the engine as CBR shows it is a recognised paradigm, not ad hoc, and its "this resembles these past cases" explanations are naturally intelligible to a non-expert.

2.9 Existing Tools for the Retail Investor

What can a retail investor already use, and what is missing?

Three kinds of tool are common today, and each leaves a gap. *Price alerts*, in every brokerage app, fire when a price or percentage threshold is crossed; they ride the same attention behaviour CLARION targets (Barber and Odean 2008), but a fixed threshold ignores each stock's own volatility (firing constantly on volatile names, rarely on calm ones) and says only *that* a level was crossed, never *why*. *News and sentiment apps* aggregate headlines and attach a sentiment score, building on the link between tone and markets (Tetlock 2007) and on finance lexicons (Loughran and McDonald 2011); but a single polarity is thin, often proprietary, and rarely connects a headline to concrete historical precedents and their outcomes. *Robo-advisors*, the most studied retail-fintech tool, genuinely help by improving diversification and curbing biases (Cardillo and Chiappini 2024; D'Acunto, Prabhala, and Rossi 2019), but they *act for* the investor with proprietary logic, the opposite of an explanation-first tool that leaves the decision to the user. Table 2.4 positions all three against CLARION.

Table 2.4: Existing retail tools versus the system proposed in this work.

Tool	What it does	Explains why?	Shows precedents?	Acts for user?
Brokerage price alert	Fires on a fixed price / % threshold	No	No	No
News / sentiment app (Tetlock 2007)	Aggregates news, scores sentiment	Partly (opaque score)	No	No
Robo-advisor (D'Acunto, Prabhala, and Rossi 2019)	Allocates and rebalances a portfolio	Rarely (proprietary)	No	Yes
This work (CLARION)	Explains events, shows evidence	Yes, by design	Yes (measured impact)	No

For CLARION: each existing tool covers one corner, alerting, news, or allocation, but none combines volatility-aware detection, transparent explanation and concrete historical evidence in one tool that informs rather than decides. That gap is the niche this work fills.

2.10 Comparative Analysis and Positioning

So what does CLARION choose, and why?

Table 2.5 gathers the choices against their alternatives. The rule throughout is *defensible simplicity* (Rudin 2019): when two options perform comparably, prefer the more transparent one, because the system must be understood and acted upon by a non-expert.

Table 2.5: Component choices in this work versus alternatives.

Component	**Chosen**	**Alternative**	**Why**
Anomaly detection	z-score (rolling)	Isolation Forest, deep detectors	Transparency; low-dimensional signal
News retrieval	Sentence-BERT + cosine	Lexicons, word2vec, LLMs	Contextual yet cheap and reproducible
Impact measurement	Event-study windows	—	Standard, reproducible, no forecasting
Explanation	Transparent design + precedents	LIME / SHAP only	Faithful by construction

2.11 Chapter Conclusions

None of these building blocks is new. What is scarce, and what this work builds, is their disciplined integration into a single explainable system for the retail investor: a transparent detector, semantic retrieval paired with event-study impact, and explanations that are faithful by construction. The next chapter turns these choices into a method.

Chapter 3

Methods and Materials

This chapter answers two questions: how is the system built, and how is it judged? It covers the approach, the data (in a reproducible data card), the techniques behind each component, the responsible-AI commitments, and the evaluation design, which was fixed before any experiment was run. One idea runs through all of it: keep things as simple as they can defensibly be, so that when two options perform about the same, the more transparent one wins. CLARION itself is described in Chapter 4; the experiments are reported in Chapter 5.

3.1 Approach and Methodology

How was the system built? Incrementally, end to end. Rather than finishing each component in isolation, a thin slice was built and proven first, from data acquisition to a delivered alert, so that integration risk surfaced early. Each component then started in its simplest defensible form and grew only where the extra complexity earned its place. This is what Artificial Intelligence *engineering* means here: the building blocks already exist in the literature (Chapter 2), so the contribution is their disciplined integration, application and critical evaluation, not a new algorithm.

Two habits follow. First, every component is defined by its *responsibility* and its *interface*, not by a particular implementation, which keeps it testable and swappable (one embedding model for another, say). Second, the project's non-negotiable constraints, only free and reproducible resources, the US market, transparency, and no prediction or trading, filter the design space before any performance question.

3.2 Datasets and Data Sources

What data does the system use? Two clearly separated layers that share one schema, so they are directly comparable (Figure 3.1): a **historical** layer, built offline, that supplies the precedents, and a **live** layer that supplies real-time prices and news. The split is deliberate: the historical layer must be rich and stable, the live layer timely and cheap.

3.2.1 Historical Layer: the FNSPID Corpus

The intended primary source for the historical layer is the FNSPID dataset (Dong, Fan, and Peng 2024), which provides financial news time-aligned with daily prices for thousands of US companies. That alignment is exactly what is needed to relate news to subsequent price impact without custom scraping. The dataset's news component is large (on the order of

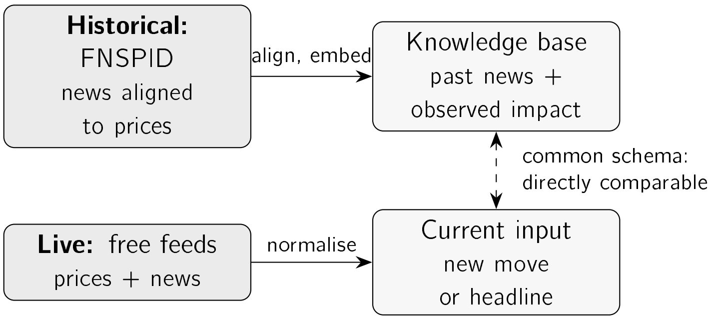

Figure 3.1: The two data layers. The historical layer turns the FNSPID corpus into a knowledge base of past news and its observed price impact; the live layer normalises real-time prices and news into the same fields, so a current event can be compared directly against the precedents.

tens of gigabytes), which directly motivates the streaming, filtered ingestion described below. Table 3.1 records the data card: the choices that make the historical subset reproducible.

Table 3.1: Data card for the *designed* historical layer (FNSPID).

Item	**Value**
Source	FNSPID (Dong, Fan, and Peng 2024) (financial news time-aligned with prices)
Licence	Creative Commons BY-SA 4.0 (attribution required)
Fields used	date, ticker, headline (mapped to a common schema)
Tickers (15)	AAPL, MSFT, AMZN, GOOGL, NVDA, TSLA, META, JPM, BAC, XOM, CVX, JNJ, PFE, WMT, KO
Sectors (5)	Technology, banking, energy, health, consumer staples
Time window	2018–2023 (daily granularity)
Prices	Daily closing prices for the same tickers
Governance	Large data not versioned (recreated by script); only small samples committed

Note: this card documents the *intended* FNSPID-based historical layer. The retrieval study of Chapter 5 is run on the real recent-news evaluation corpus described in Section 3.2.3 (3,714 headlines for the same fifteen tickers, built through the identical process); the full multi-year FNSPID build is identified as future work.

The fifteen large-cap tickers span five sectors, which makes the cross-sector evaluation of Chapter 5 meaningful while keeping the corpus tractable; the 2018–2023 window (the dataset's extent) is recent yet long enough to hold precedents. These are documented choices, not fixed ones: a different ticker set or window can be substituted without changing the method.

3.2.2 Preprocessing and Event Alignment

How is a news item turned into a historical case? In three steps. First, the news file is read in a stream and filtered to the chosen tickers and window, so only the relevant subset is materialised. Second, each item is normalised to the common schema (date, ticker, headline). Third, each item is aligned to the market and its impact measured. The alignment rule matters for honesty: the event day is the first trading day *on or after* the news date, and impact is measured from that day's *close* onward. Measuring from the close, not the open, avoids crediting a move already in the opening price, and measuring strictly forward enforces the no-lookahead guarantee of Section 3.6. Items with no aligned prices, or whose window runs past the data, are dropped rather than imputed. Algorithm 3.1 states the build.

Algorithm 3.1 Knowledge-base construction with event alignment.

Require: news items $\{(d_i, c_i, h_i)\}$; daily close prices P_c per ticker; horizons $H = \{1, 3, 5\}$
Ensure: knowledge base $\mathcal{K}$ of cases $(d, c, h, \mathbf{e}, \text{impact})$

$\mathcal{K} \leftarrow \emptyset$
for each news item (d_i, c_i, h_i) **do**
  $e_i \leftarrow$ first trading day $\geq d_i$ in P_{c_i} ▷ event alignment
  if e_i exists **then**
    $\mathbf{e}_i \leftarrow \text{embed}(h_i)$ ▷ Sentence-BERT
    for $\tau \in H$ **do**
      $\text{impact}_i[\tau] \leftarrow$ return from close(e_i) to close($e_i+\tau$), or n/a if unavailable
    end for
    $\mathcal{K} \leftarrow \mathcal{K} \cup \{(d_i, c_i, h_i, \mathbf{e}_i, \text{impact}_i)\}$
  end if
end for
return $\mathcal{K}$

3.2.3 Live Layer and Evaluation Dataset

The live layer provides real-time prices from a free market-data service (with a free fallback) and news from a free company-news service and RSS feeds. Free news is deliberately *not* used to build the historical layer, because free tiers cover only a short, recent, delayed window; but it is fine as the live trigger and as a source of *real* news for a first evaluation. For that, a set of 3,714 real headlines for the fifteen tickers was collected from the free company-news service (the most recent months available), and is used for the retrieval study in Chapter 5. This corpus is real but recent, which is why the full multi-year FNSPID build remains the richer source, identified as future work.

3.2.4 Data Governance and Attribution

Two governance rules apply to both layers. First, large data and models are never versioned: they are recreated by the ingestion process, and only small illustrative samples are kept under version control. Second, third-party news text is not republished: the dataset is cited and attributed as required by its Creative Commons BY-SA 4.0 licence (Dong, Fan, and Peng 2024), and only minimal headline excerpts appear in this document.

3.3 Models and Techniques

Three techniques do the work: a detector, a retrieval-and-impact engine, and an explanation step. Each is introduced by what it is *for*, then how it works.

3.3.1 Statistical Anomaly Detection

What it is for: to flag a daily move that is unusual *for that stock*, not just large in absolute terms. *How it works:* let r_t be the day's (log-)return and μ_t, σ_t the mean and standard deviation of returns over a trailing window of w days ending strictly before t. The day is flagged when the absolute z-score exceeds a threshold k:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \qquad \text{anomaly} \iff |z_t| > k. \tag{3.1}$$

This is the statistical family of detectors (Chandola, Banerjee, and Kumar 2009), chosen for transparency: the investor can be shown exactly how many standard deviations the move was. More expressive detectors exist, Isolation Forest (F. T. Liu, Ting, and Zhou 2008) or deep models (Pang et al. 2021), but their opacity is not warranted for a single return series explained to a non-expert (Chapter 2). The window w and threshold k are explicit choices, examined by ablation in Chapter 5. Algorithm 3.2 states the rule; the window for day t uses only earlier days, so no future information enters.

Algorithm 3.2 Rolling z-score anomaly detection (no lookahead).

Require: return series $r_1, \ldots, r_T$; window w; threshold k
Ensure: flag a_t for each day, with the quantities behind it

```
for t ← w + 1 to T do
    W ← (r_{t−w}, ..., r_{t−1})                              ▷ strictly earlier days
    μ_t ← mean(W);   σ_t ← std(W)
    z_t ← (r_t − μ_t) / σ_t
    a_t ← (|z_t| > k)
end for
return for each flagged t, the tuple (z_t, μ_t, σ_t, w, k)   ▷ exposed to the explanation engine
```

A small example shows why this matters. Two stocks both fall 3.2% today. One has been calm ($\sigma \approx 0.40\%$), the other volatile ($\sigma \approx 1.50\%$). The same move is a clear anomaly for the calm stock ($z \approx -8.1$) but an ordinary day for the volatile one ($z \approx -2.2$), as Table 3.2 shows. A fixed percentage threshold would flag both; the z-score does not, which is exactly what we want.

Table 3.2: Worked example: the same −3.2% move on a calm and a volatile stock.

Quantity	**Calm stock**	**Volatile stock**
Recent mean μ	0.04%	0.10%
Recent standard deviation σ	0.40%	1.50%
Today's return r	−3.2%	−3.2%
z-score $(r-\mu)/\sigma$	**−8.1**	**−2.2**
Flagged at $k = 3$?	yes	no

3.3.2 Semantic Retrieval and Impact Measurement

What it is for: given a new headline, find the most similar past headlines and show what happened to their prices. This news–market correlation engine is the heart of the work, and it combines two techniques. *Retrieval:* each headline becomes a sentence embedding with Sentence-BERT (Reimers and Gurevych 2019), chosen (Section 2.5) because it captures meaning yet yields directly comparable vectors, so the nearest past items are found cheaply by cosine similarity. *Impact:* for each precedent, the cumulative return is measured over fixed post-event windows (+1, +3 and +5 trading days), in the manner of an event study (Brown and Warner 1985; Fama et al. 1969). Both the similarity metric and the windows are explicit, documented choices, and together they are the methodological contribution. Impact is measured strictly after the event, so it is an outcome, not a forecast. Algorithm 3.3 states the retrieve-and-reuse step (the case-based-reasoning core of Section 2.8).

Algorithm 3.3 Precedent retrieval and impact reporting (news trigger).

Require: headline h; knowledge base $\mathcal{K} = \{(d_j, c_j, h_j, \mathbf{e}_j, \text{impact}_j)\}$; neighbours k; horizon τ
Ensure: top-k precedents with similarity and observed impact, and their mean impact

```
q ← embed(h)                                   ▷ same Sentence-BERT model as the KB
for each case j ∈ K do
    s_j ← cos(q, e_j)                          ▷ cosine over L2-normalised vectors
end for
S ← indices of the k largest s_j
return {(d_j, c_j, h_j, s_j, impact_j[τ]) : j ∈ S} and mean_{j∈S} impact_j[τ]
```

The impact itself is read off a short window after the event, as Figure 3.2 shows. The news may arrive on a non-trading day, so the *event day* is the first trading day on or after it; the return is then accumulated from that day's close to the close +1, +3 and +5 trading days later. Anchoring on the close, and looking only forward, is what keeps the measure free of lookahead: it records what happened *after* the event became actionable, never before.

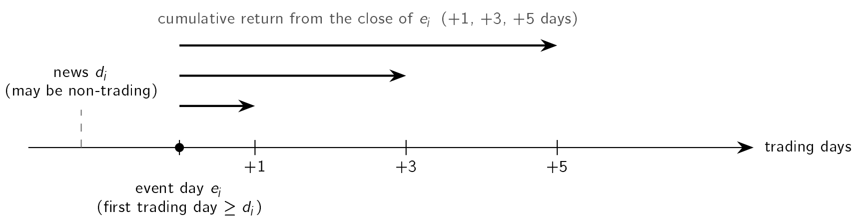

Figure 3.2: How impact is measured. The news date d_i may fall on a non-trading day; the event day e_i is the first trading day on or after it. The return is then accumulated forward from the close of e_i over +1, +3 and +5 trading days. Only post-event days are used, so no future information enters.

Three choices fix *how* impact is measured, each made for transparency as much as rigour:

- **Horizons.** Short windows (+1, +3, +5 days), standard for daily event studies (Brown and Warner 1985); longer windows let unrelated, market-wide moves creep in.
- **Baseline.** The *raw* cumulative return, not a market-adjusted abnormal return, because it is directly observable and needs no estimated market model a non-expert could not check. The cost, stated plainly, is that a raw return mixes the news reaction with any market-wide move on the same days (the confounding threat examined in Chapter 5); market adjustment is future work.

- **Aggregation.** The headline figure is the *mean* impact across the retrieved precedents, but the individual precedents are always shown too, so the investor sees the spread, not just one number.

A worked example makes it concrete. Figure 3.3 gives the intuition: the embedding places each headline in a space where similar themes fall close together, so a query lands near its matches and far from unrelated news.

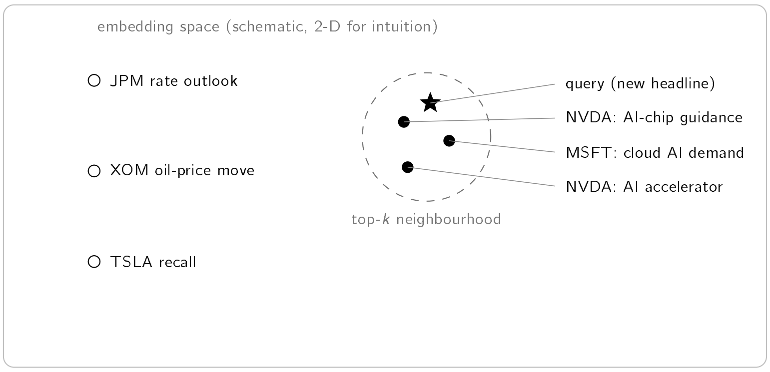

Figure 3.3: Conceptual illustration of semantic retrieval. Each headline becomes a point in a high-dimensional embedding space (drawn in two dimensions for intuition only). Texts on the same theme lie close together, so a query (star) falls near its semantically similar precedents (filled), which form the top-k neighbourhood, and far from unrelated news (hollow). The next table makes this concrete with real numbers.

Table 3.3 runs the step end to end on the small sample knowledge base shipped with the system. To keep it reproducible without downloading a model, it uses a transparent word-overlap encoder (the deployed system uses Sentence-BERT, evaluated in Chapter 5). For the query *"Nvidia demand surges on AI chip orders"*, the three nearest precedents are all AI-themed and not all from the same company: a Microsoft cloud-AI item is retrieved alongside two Nvidia items, showing the *cross-ticker* generalisation the evaluation rests on. Their mean +5-day impact is +6.5%, shown as evidence, with the explicit caveat that it records what followed comparable past news, not a forecast.

Table 3.3: Worked retrieval example on the bundled sample knowledge base. Query: *"Nvidia demand surges on AI chip orders"*; similarities and impacts are real and reproducible.

Similarity	**+5-day impact**	**Ticker / date**	**Retrieved headline (excerpt)**
0.60	+3.5%	NVDA, 2023-05-25	Nvidia guidance surges on soaring demand for AI data-centre chips
0.38	+10.9%	MSFT, 2023-04-25	Microsoft cloud growth accelerates on AI demand
0.38	+4.9%	NVDA, 2023-06-13	Nvidia unveils new AI accelerator to extend data-centre lead
Mean +5-day impact across the retrieved precedents: +6.5%			

Note: the mean is computed from the unrounded impacts, so it need not equal the average of the rounded values shown.

3.3.3 Explanation Techniques

What it is for: to turn the computed objects into a message the investor can check. Explanation is pursued by transparent design (Section 2.3) (Barredo Arrieta et al. 2020; Rudin 2019). Each alert is assembled from three transparent ingredients: the rule that fired (the z-score with its window and threshold); the retrieved precedents with their observed impact; and, optionally, feature attribution over the detector with SHAP (Lundberg and S.-I. Lee 2017). Financial sentiment from a domain model (Araci 2019) is an optional enrichment, used only if it adds defensible value. Because every alert is rendered directly from these objects, the explanation is faithful to the computation by construction, a property tested in Chapter 5.

3.4 Tools and Frameworks

The system is built on the open-source scientific-Python ecosystem: mature, free libraries for numerical computing, market-data access, sentence embeddings and figures. This fits the free-resources constraint and reproducibility, since pinned dependencies and fixed seeds let the environment and results be recreated elsewhere. The concrete engineering (environment, code organisation, tests) is kept in Appendix A, so this chapter stays a description of *methods*, not software.

3.5 Responsible AI and Ethics

The work is bound by several responsible-AI commitments that follow directly from its purpose and its audience. **No advice and no prediction:** the system performs neither price prediction nor trading recommendation; it surfaces observed past outcomes as evidence and states this explicitly in every alert, so that the investor is informed rather than instructed. **Transparency:** every alert exposes its full reasoning chain, in keeping with the explainability principles of Chapter 2, so that a user can verify rather than merely trust the output. **Honest evidence:** only results that can be reproduced and explained are reported, and no data, metric or citation is fabricated. **Data stewardship:** all sources are used within their free tiers and licences, the FNSPID dataset is attributed as its Creative Commons licence requires (Dong, Fan, and Peng 2024), third-party news text is not republished, and no personal data is processed. **Security:** credentials are never stored in versioned files. Finally, the system is positioned as a decision-support aid for a non-professional investor, not as a substitute for professional advice, a limitation stated plainly rather than hidden.

3.6 Evaluation Methodology

How is the system judged? By a plan fixed before any experiment ran, so the metric cannot be chosen after the fact to flatter the result, a small-scale pre-registration. The results are in Chapter 5; here is the protocol they follow.

3.6.1 Retrieval metric and relevance

Retrieval is evaluated as an information-retrieval task (Section 2.6) with precision@k, the share of the top k precedents that are relevant. Writing Q for the set of query headlines and $R_k(q)$ for the top k items returned for query q,

$$\text{precision@}k = \frac{1}{|Q|}\sum_{q\in Q}\frac{1}{k}\sum_{d\in R_k(q)}\text{rel}(q,d), \qquad \text{rel}(q,d) = \mathbf{1}\big[\,\text{sector}(d) = \text{sector}(q)\,\big]. \tag{3.2}$$

Relevance is approximated by *sector agreement*: a precedent counts as relevant if its company is in the query's sector. This is an automatic stand-in for "analogous", used because no labelled set exists; its limits are stated with the results. Two choices make the metric demanding, not flattering. First, a strict *cross-ticker* constraint drops every precedent sharing the query's ticker, so the system cannot win by matching a company to its own past news; it must generalise *across* companies in a sector. Second, k is small (the headline figure uses $k = 5$), reflecting that an alert shows only a few precedents.

3.6.2 Baselines

A single number means little alone, so semantic retrieval is compared against three baselines, each controlling for a rival explanation. **Random** draws k precedents at random; its precision is the sector base rate, the no-skill floor. **Recency** returns the newest items, testing whether simply surfacing fresh news would do as well. **Lexical** ranks by word overlap under the same cosine, so the gap to the embedding model isolates the value of *meaning* over surface words. Each comparison is repeated over five random seeds (mean with standard deviation), and two Sentence-BERT variants are tried, so the result reflects semantic retrieval in general, not one lucky draw or one model.

3.6.3 Anomaly-detection protocol

The anomaly detector is judged primarily by an argument that needs *no label*: a good universal detector should fire at a comparable rate across instruments of very different volatility, whereas a volatility-blind rule cannot. Consistency is therefore measured as the range (maximum minus minimum) of per-ticker firing rates: the smaller the range, the more uniform the detector. Only secondarily is the detector scored against an explicit *extreme-move* proxy (a daily move at or beyond the 99th percentile of that ticker's own returns), via precision, recall and F_1; because this proxy is itself volatility-relative, it is treated as supporting rather than primary evidence. An ablation over the estimation-window length completes the picture by exposing the detector's sensitivity to its one main parameter.

3.6.4 Explanation, scope, and rigour guarantees

The explanation engine is assessed for *fidelity*: whether the alert text reflects the system's actual computation. This holds by construction, because each alert is rendered directly from the computed objects. Its *usefulness* to a real investor is a separate matter, which would need a human study not yet conducted, and is reported as a limitation rather than a claim. Because the system performs neither price prediction nor trading, profit-based metrics are out of scope by design, not omitted by convenience. Underpinning every experiment are three guarantees. First, **no lookahead**: any feature computed at an instant uses only information available at or before that instant, and historical impact uses only post-event windows. Second, **reproducibility**: random seeds are fixed, dependencies are pinned, and every data and results figure is produced by a versioned script. Third, **honesty**: results are reported with their caveats, and modest but genuine findings are preferred to inflated ones.

3.7 Chapter Conclusions

This chapter set the method: defensible simplicity and incremental delivery; a two-layer data architecture in a reproducible data card; the techniques for detection, retrieval, impact and explanation, each introduced by its purpose; and an evaluation fixed in advance. The next chapter assembles these into CLARION.

Chapter 4

CLARION: An Explainable Financial-Alert System for Retail Investors

This chapter presents Clarion, the system built in this dissertation: its architecture, the historical knowledge base and correlation engine at its core, the anomaly trigger, the explanation engine that makes every alert traceable, and the way alerts reach the investor. Chapter 3 argued for the individual methods; the focus here is how they fit together into one working system.

4.1 Overview

Clarion watches the US equity market on behalf of a retail investor and raises an alert whenever one of two independent triggers fires: an abrupt price movement, or a relevant incoming news item. Each alert is accompanied by a full, traceable explanation rather than a bare notification. The system is organised as a pipeline of well-separated components, each with a single responsibility, connected by a common data schema so that the live and historical layers are directly comparable.

4.2 Architecture

The system is driven by the two triggers and converges on a single explanation step before delivery. Figure 4.1 shows the components and the flow of data between them. The live layer supplies real-time prices and news, and the historical knowledge base supplies the precedents. The anomaly detector and the correlation engine turn these into evidence, the explanation engine assembles a traceable account, and the delivery component sends the alert to the investor.

Three engineering principles recur throughout and make the system both testable and defensible. First, **a thin end-to-end slice was built before breadth**: the market-movement trigger was implemented and proven end to end (from price retrieval to a real alert) before the heavier correlation engine was added, reducing integration risk. Second, **pure logic is separated from input/output**, so the numerical core can be verified deterministically and offline, without a network connection or the heavy model stack. Third, **components are programmed against interfaces**: the embedding step is defined abstractly, with two

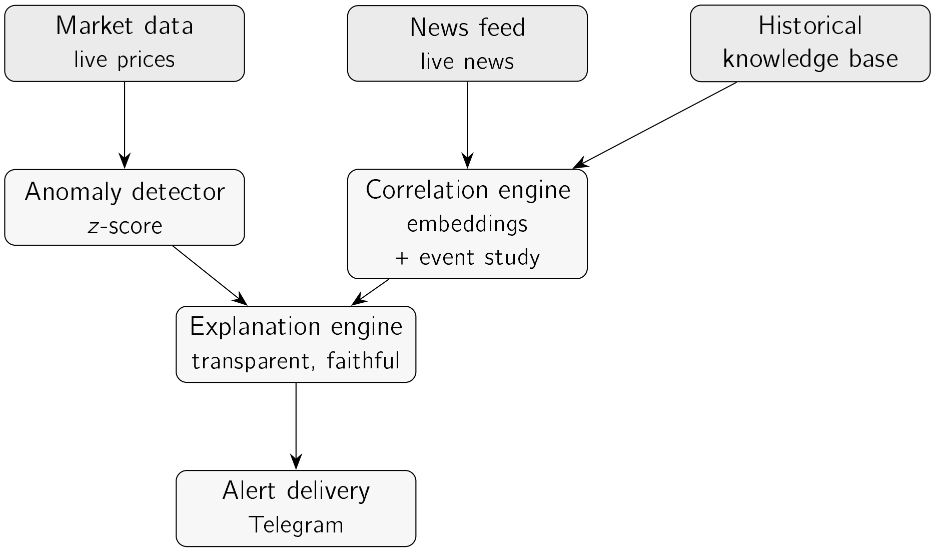

Figure 4.1: Architecture of Clarion. The live layer (market data, news feed) and the historical knowledge base feed two engines (the anomaly detector and the correlation engine) whose evidence converges on the explanation engine before an alert is delivered through Telegram. The market-movement trigger follows the left path; the news trigger the right.

interchangeable implementations (a dependency-free lexical baseline and the real Sentence-BERT model), so that one can be substituted for the other without changing anything else, which is what makes the embedding-model comparison in Chapter 5 a fair one.

Table 4.1 gathers the principal design decisions and the rationale behind each, making explicit the engineering judgement that, in a work of this kind, *is* the contribution. Every decision follows the guiding principle of defensible simplicity established in Chapter 3.

Where Figure 4.1 shows the components, Figure 4.2 traces the same system as one connected flow of data and processing steps: the offline knowledge-base build (top), the news trigger (middle) and the market trigger (bottom), all converging on the explanation engine and the alert.

4.3 Historical Knowledge Base and Correlation Engine

The historical layer is a knowledge base of past news, each entry storing its date, ticker, headline, sentence embedding and measured post-event impact. Each news item is aligned to the market by taking the first trading day on or after the news date and measuring impact only from that day's close onward; this is the no-lookahead rule made concrete, and it avoids crediting the system with a jump already embedded in the opening price. The intended primary source is the FNSPID corpus (Dong, Fan, and Peng 2024); because building the full multi-year corpus is a long one-off job, the evaluation in Chapter 5 uses an initial knowledge base built from real recent news through the identical build process, and the full build is identified as future work.

The correlation engine has two parts. Similarity is computed with cosine over the (normalised) embeddings, returning the nearest historical items to a query. Impact is computed as the cumulative return at each horizon (+1, +3, +5 days) measured from the event

Table 4.1: Principal design decisions in Clarion and their rationale.

Decision	**Choice**	**Rationale**
Detection method	Rolling z-score	Transparent; every alert is explainable to a non-expert
News representation	Sentence-BERT embeddings	Contextual meaning, yet cheap and reproducible to compare
Impact measurement	Event-study windows	Standard, reproducible; characterises an outcome, not a forecast
Explanation strategy	Transparent design + precedents	Faithful by construction, not a post-hoc approximation
Data architecture	Shared schema, two layers	Live and historical evidence are directly comparable
Component coupling	Programmed to interfaces	Testable offline; enables fair model substitution
Delivery channel	Telegram bot	Free, ubiquitous, no bespoke client needed

close, with horizons that fall outside the available series reported as not available rather than silently dropped. Given a new item, the engine embeds it, ranks the knowledge base by similarity, and returns the top precedents with their scores, the evidence later shown to the user. The middle lane of Figure 4.2 traces these steps end to end, from an incoming headline to the delivered alert: the headline is embedded, matched against the knowledge base by similarity, and its nearest precedents are returned with their measured impact, ready for the explanation engine. The dashed precedents arrow makes plain that the knowledge base is consulted but never decides; the reasoning stays in the open.

4.4 Anomaly Trigger

The anomaly trigger applies Equation 3.1 directly. For the day under evaluation it computes the mean and standard deviation over the window of strictly preceding days, forms the z-score of the latest return, and returns the decision together with every quantity that produced it: the z-score, window, threshold, mean and standard deviation. Exposing these values is what allows the explanation engine to be faithful by construction, and the evaluation applies exactly this rule across entire price series so that the experiments use the same logic as production. The bottom lane of Figure 4.2 traces the market trigger from a daily price to the delivered alert: the latest return is standardised against its rolling window, and the move, with all the quantities behind it, is passed on only when the absolute z-score exceeds the threshold.

4.5 Explanation Engine

The explanation engine renders alerts directly from the objects produced upstream, which guarantees that the text never diverges from the computation. For the market trigger, it states the move, the z-score, the threshold and window, and the recent mean and standard

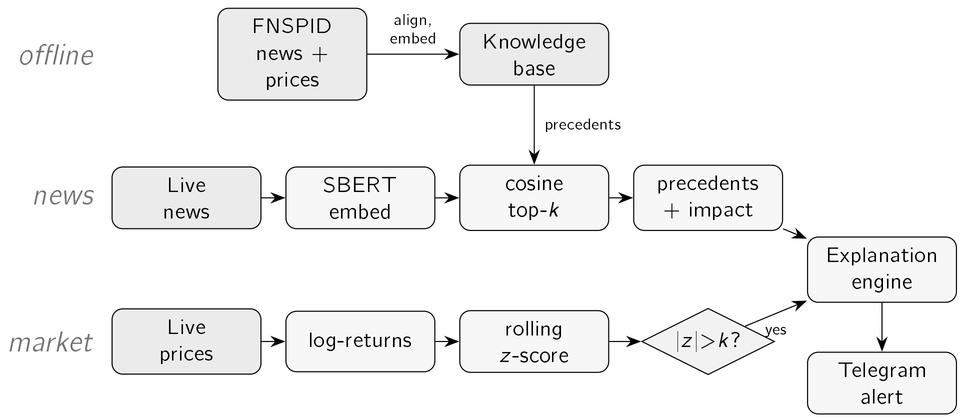

Figure 4.2: The end-to-end flow of Clarion as one connected pipeline. *Offline* (top), FNSPID news is aligned to prices and embedded into the knowledge base. The *news trigger* (middle) embeds an incoming headline, finds its nearest precedents in the knowledge base and attaches their observed impact; the *market trigger* (bottom) turns live prices into a z-score and fires only when it exceeds the threshold. Both paths converge on the explanation engine, which delivers one explained alert through Telegram.

deviation, and it renders the z-score in plain language (how many times a typical day's move it represents), so that a non-expert can interpret the alert rather than merely see the numbers. For the news trigger, it lists the retrieved precedents (date, ticker, similarity score, measured impact and headline) and the average impact across them, and it always ends with an explicit disclaimer that the figures are observed past outcomes, not a prediction, keeping the system within its no-forecasting scope.

4.6 Alert Delivery

Alerts are delivered to the investor through the Telegram messaging platform, chosen because it offers a free, widely used bot interface. The two end-to-end flows (retrieve, detect, explain and send for the market trigger; embed, retrieve precedents, explain and send for the news trigger) each encapsulate a complete run; periodic scheduling and deployment are deliberately left outside the present scope. A real alert was successfully delivered during testing; Figure 4.3 reproduces the format the investor receives, with the full reasoning chain visible in a single message.

4.7 Deployment and Practical Use

CLARION runs as two entry points, one per trigger, each of which performs a complete cycle. The market trigger fetches recent prices, computes returns, detects, explains and sends; the news trigger takes a headline, embeds it, retrieves precedents, explains and sends. In day-to-day use these would be invoked on a schedule (the market trigger once per trading day after the close, the news trigger whenever a watched feed publishes), but the scheduler and the hosting fall outside the project's scope. The entry points already encapsulate everything such a scheduler would call.

Running the system needs three things, all free: a market-data source for prices, a company-news or RSS source for headlines (an API key, kept out of version control), and the historical

CLARION — financial alerts bot

News alert — NVDA

"Nvidia raises guidance on surging demand for AI data-centre accelerators"

Potential impact (across the 3 precedents shown): average 1-day move −2.2%

Historical precedents:
- MSFT (sim 0.68) +1d −3.5%
- META (sim 0.68) +1d −2.7%
- GOOGL (sim 0.64) +1d −0.5%

Precedents are retrieved by semantic similarity; the impact is the observed past outcome, not a price prediction.

Figure 4.3: A CLARION news alert as delivered through Telegram. A single message carries the full, traceable reasoning chain: the detected event, the average observed impact of analogous past events, the individual precedents with their similarity scores and measured impact, and an explicit no-prediction disclaimer. The figures shown are a representative, rounded illustration; the full real end-to-end output appears in Chapter 5 (Case Study 3).

knowledge base, built once from the corpus. Appendix A gives the exact steps. The real alert delivered during testing shows that the path from raw data to the investor's phone works end to end, not just on paper.

Two practical limits are worth stating plainly. First, the free news tier returns only a recent window, so the knowledge base is shallower than the multi-year build it is designed for; this affects coverage, not the method. Second, CLARION is a decision-support prototype rather than a production service: it has no user accounts, no persistence beyond the knowledge-base file, and no automatic recovery from a provider outage. None of these is a research limitation; each is a known, ordinary step on the road from prototype to product, and is recorded here openly rather than hidden.

Beyond these engineering gaps, an alerting tool aimed at non-experts raises two responsible-design questions that the prototype surfaces rather than solves. The first is alert fatigue: a tool that fires too often trains the user to ignore it, defeating its purpose. A production version would rank alerts by severity (for the market trigger, the magnitude of the z-score) and rate-limit the least important notifications, so that attention is reserved for genuinely unusual events. The second is over-reliance: a confident-looking explanation can be trusted more than its evidence warrants, the failure mode reviewed in Section 2.4. The present design mitigates this by showing the individual precedents and their spread rather than a lone averaged number, and by disclaiming prediction on every alert; natural extensions are to de-duplicate near-identical precedents and to flag when a retrieved cluster disagrees in direction, so the user sees genuine uncertainty instead of false consensus. Both are recorded as design considerations and future work, not as implemented features.

4.8 Chapter Conclusions

This chapter described CLARION at the level of its architecture and design: a two-trigger, two-layer pipeline whose components are separated by responsibility and connected by a common schema, with explainability built in by rendering every alert directly from the system's own computation. The next chapter evaluates the system's two core components and its end-to-end behaviour on real data.

Chapter 5

Case Studies

This chapter evaluates CLARION through three case studies on real data. The first looks at the anomaly trigger across stocks of very different volatility; the second at the correlation engine and the precedents it retrieves; the third walks through a complete alert and asks whether its explanation is faithful. Every number below comes from a versioned script with a fixed seed, the evaluation followed the plan set out in Chapter 3, and each study states its own limitations.

5.1 Common Experimental Setup

Two real datasets are used. For the anomaly trigger, daily closing prices for the fifteen large-capitalisation tickers listed in the data card were retrieved from a free market-data service over a fixed three-year window (June 2023 to June 2026), yielding 750 daily log-returns per ticker. For the correlation engine, 3,714 real financial-news headlines for the same tickers were collected through a free company-news service (the most recent months available on the free tier), spanning five sectors (technology, banking, energy, health and consumer staples). This is the evaluation corpus actually used here, distinct from the designed FNSPID layer documented in the data card of Chapter 3. News headlines and prices are normalised to a common schema so that the live and historical layers are directly comparable. This recent window of a few months is enough for the retrieval experiment; a full impact analysis still needs the multi-year FNSPID corpus, left as future work. The no-lookahead, reproducibility and honesty guarantees of Section 3.6 apply to every study below.

5.2 Case Study 1: Transparent Anomaly Detection on US Equities

The question is whether the rolling z-score (Chandola, Banerjee, and Kumar 2009) is a better universal detector of abrupt moves than a naïve fixed-percentage threshold. The proxy label for a "true" anomaly is a daily move in the upper percentile of *that ticker*'s own return distribution ($|r| \geq$ the 99th percentile); the baseline is a single global threshold ($|r| \geq 3\%$) applied to every ticker. This label is itself volatility-relative, so any agreement score it produces is only supporting evidence; the primary argument, made first below, is the firing-rate consistency, which needs no label at all.

Firing-rate consistency (main argument). The strongest evidence is independent of any label. A good universal detector should fire at a comparable rate across instruments; a volatility-blind threshold cannot. Table 5.1 and Figure 5.1 show that the fixed threshold fires on roughly 1% of days for a calm stock (KO) but on about 35% of days for a volatile one (TSLA, NVDA), whereas the z-score fires at a near-constant rate ($\approx$ 2–3%) for all tickers. The range of firing rates across tickers is more than twenty times tighter for the z-score (0.015 versus 0.344), confirming that normalising by recent volatility makes detection comparable across the market.

Table 5.1: Firing rate across the fifteen tickers (three years of daily data).

Method	Min rate	Max rate	Range
z-score (window 20, ± 3)	0.016	0.031	**0.015**
Fixed threshold ($\lvert r\rvert \geq 3\%$)	0.009	0.353	**0.344**

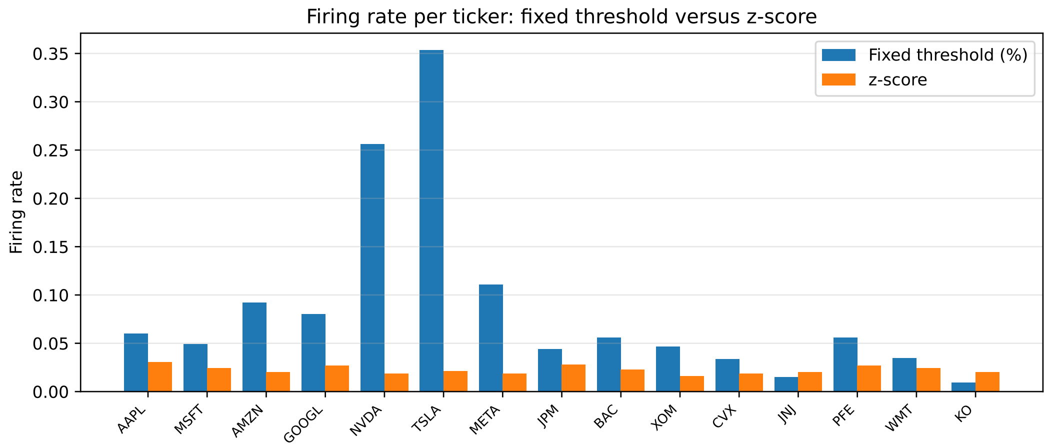

Figure 5.1: Firing rate per ticker. The fixed percentage threshold varies dramatically with each stock's volatility, while the z-score is stable across all tickers.

A worked example. Figure 5.2 shows the detector at work on a single volatile stock (Tesla) over the three-year window. The flagged days are not simply the largest raw moves: they are the moves that are large *relative to the asset's own recent volatility*, which is exactly the behaviour a fixed percentage threshold cannot reproduce. Over this period the detector flags 16 of the 750 trading days (about 2%), consistent with the near-constant firing rate reported above.

To make this concrete, consider the single largest surprise in the window. On 24 October 2024, following its quarterly earnings, Tesla's daily log-return was +0.198, a price move of about +22%. Measured against the preceding twenty days, whose mean log-return was −0.92% and standard deviation 2.73%, this is a z-score of +7.6, far beyond the threshold $k = 3$ (Table 5.2). The same rule that leaves an ordinary two-percent wobble unflagged identifies this earnings reaction at once, and exposes the quantities that justify the alert: the move, the recent norm it is judged against, and the resulting z-score, which is what the explanation engine then shows the investor.

Agreement with the proxy label (supporting). Against the per-ticker extreme-move label, the z-score attains an F_1 of 0.516 (precision 0.381, recall 0.800) versus 0.218 for the fixed threshold (precision 0.122, recall 0.992): the fixed rule catches almost every large raw

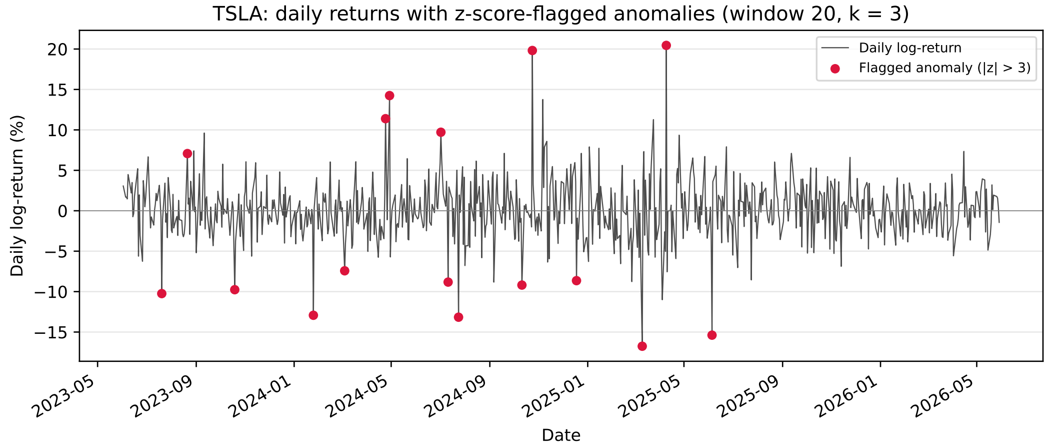

Figure 5.2: Daily log-returns of a volatile stock (Tesla) over the evaluation window, with the days flagged by the z-score detector (window 20, $k = 3$) marked. The detector responds to volatility-relative surprise rather than to absolute move size.

Table 5.2: Worked anomaly: Tesla's post-earnings move on 24 October 2024 (window 20, $k = 3$); real, reproducible figures.

Quantity	**Value**
Trailing-20-day mean of log-returns, μ	−0.92%
Trailing-20-day standard deviation, σ	2.73%
Day's log-return, r (a price move of $\approx +22\%$)	+19.82%
z-score, $(r - \mu)/\sigma$	**+7.61**
Flagged at $k = 3$?	yes

move but at very low precision. A window ablation (Figure 5.3) shows F_1 rising with the estimation window (0.385, 0.516 and 0.678 for 10, 20 and 60 days), indicating that a longer volatility estimate yields a steadier baseline before it begins to saturate. This label is itself volatility-relative, so some circularity is unavoidable; that is why the firing-rate consistency above, which depends on no label, is treated as the primary evidence.

5.3 Case Study 2: News–Market Precedent Retrieval

The central claim of this work is that semantic retrieval finds genuinely analogous precedents. This is measured as precision@k with a sector proxy, in a *cross-ticker* setting: for each query the k most similar headlines *from other companies* are retrieved, and precision is the fraction sharing the query's sector. Excluding the query's own company prevents the trivial solution of matching on the company name and tests thematic analogy instead. It is worth being clear about what this measures: same sector is an automatic stand-in for relevance, so the metric shows whether retrieved news is thematically related, not whether a person would judge it a useful precedent. That gap is real, and a human relevance study is left as future work; the proxy's merit is that it is objective and reproducible, which is what a first evaluation needs. To check that the result is not tied to one model, *two* Sentence-BERT models (Reimers and Gurevych 2019) (the lightweight default, MiniLM, and the larger MPNet) are compared

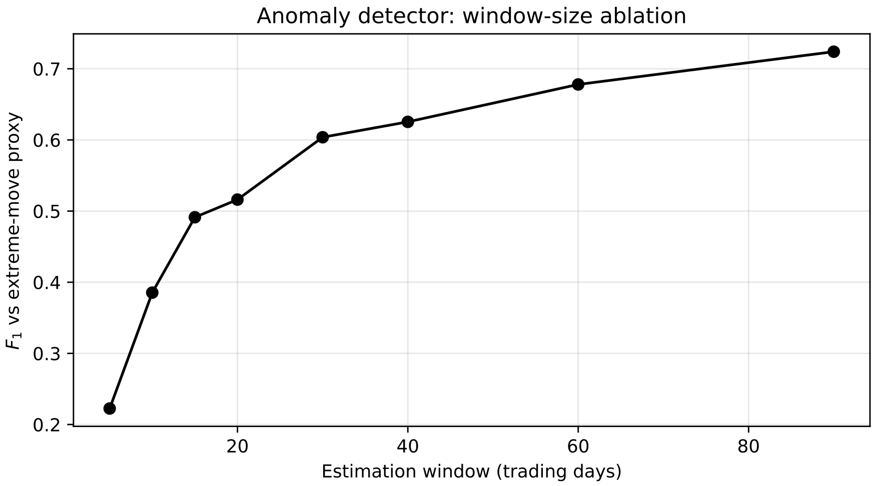

Figure 5.3: Effect of the estimation-window length on the detector's F_1 against the extreme-move proxy. A longer window yields a steadier volatility baseline and higher agreement, with diminishing returns at the longest windows.

against a lexical (hashing) baseline and two trivial baselines: random retrieval (the sector base rate) and recency. To guard against sampling artefacts, five hundred queries were sampled in each of five independent runs (seeds 42–46); Table 5.3 reports the mean and standard deviation.

Table 5.3: Cross-ticker precision@k by sector, mean ± std over five runs (3,714 real headlines, five sectors).

Method	**P@5**	**P@10**
SBERT (MiniLM, default)	0.514 ± 0.015	0.478 ± 0.011
SBERT (MPNet)	$\mathbf{0.538 \pm 0.011}$	$\mathbf{0.506 \pm 0.010}$
Lexical baseline (hashing)	0.346 ± 0.011	0.314 ± 0.008
Recency	0.126 ± 0.006	0.106 ± 0.005
Random (base rate)	0.240 ± 0.004	0.240 ± 0.004

As Table 5.3 and Figure 5.4 show, the default MiniLM model reaches a precision@5 of 0.514, about 2.1 times the random base rate (0.240) and well above the lexical baseline (0.346). The stronger MPNet model does slightly better (0.538), which suggests the advantage belongs to semantic embeddings in general rather than to one particular model; the small standard deviations (around 0.01) show the gap is robust rather than a sampling artefact. Recency performs worst, below even the random base rate, so simply surfacing the latest news is no substitute for semantic matching. The lift over both the random and lexical baselines is the honest measure of the value added by the embeddings.

Impact measurement. For each retrieved precedent the engine reports the observed post-event return at +1, +3 and +5 trading days, measured from the event-day close in the style of an event study (Brown and Warner 1985). These figures are observed outcomes presented as evidence, never a prediction. On the worked knowledge base the measured impacts are directionally coherent (for example, the competitive-threat news cluster in the end-to-end

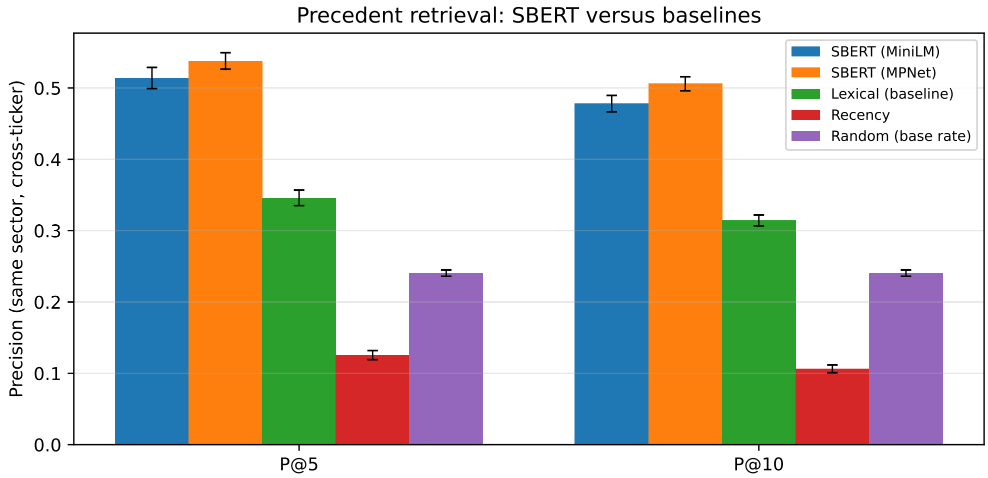

Figure 5.4: Cross-ticker precision@k by sector. SBERT clearly exceeds the lexical, recency and random baselines; error bars show ± 1 standard deviation over the five runs.

case study below is followed by uniformly negative one-day returns), which provides face validity for the measurement; a rigorous multi-year impact study on the full FNSPID corpus is identified as future work.

Retrieval by sector. The aggregate figure hides useful variation across sectors. Table 5.4 and Figure 5.5 break the retrieval down by the query's sector, using the whole population of each sector as queries (so the figures are exact rather than sampled). Two readings matter. The raw precision@5 is highest for technology (0.712), but largely because technology dominates the corpus (47% of headlines) and therefore has a high random base rate (0.429); the fair cross-sector measure is the *lift* over that base rate. By that measure the engine adds the most value in **energy** (+0.377) and **health** (+0.348), sectors whose vocabulary is distinctive (production, output, barrels; trial, drug, approval), and the least in **consumer** (+0.100), whose headlines are more generic and thematically overlap with other sectors. Retrieval is positive in every sector, but its benefit is greatest where the domain language is most specific.

Table 5.4: Cross-ticker precision@k per sector (whole-population queries, MiniLM).

Sector	N	P@5	P@10	Random	Lift P@5
Technology	1736	0.712	0.675	0.429	+0.283
Energy	497	0.448	0.408	0.072	**+0.377**
Health	494	0.419	0.379	0.071	+0.348
Banking	496	0.272	0.228	0.072	+0.200
Consumer	491	0.171	0.150	0.071	+0.100

Note: the lift is precision@5 minus the random base rate, computed from unrounded values; it may differ from the rounded columns by ± 0.001.

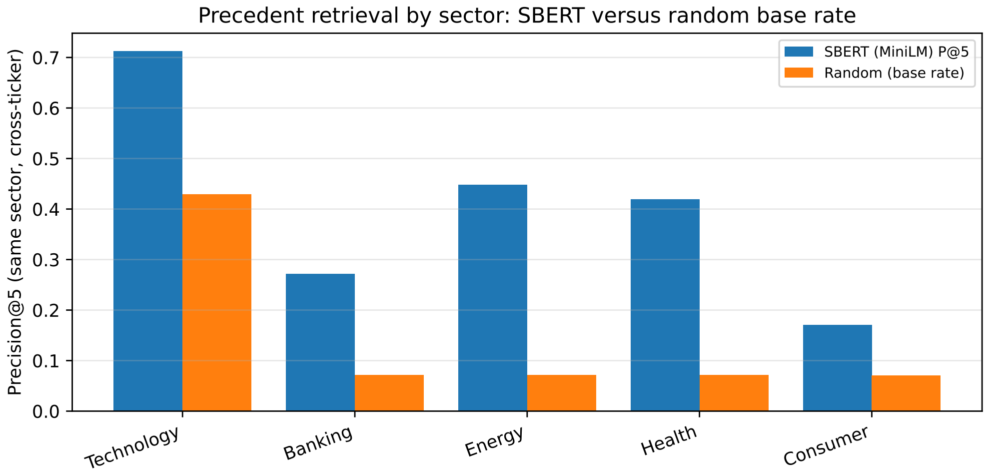

Figure 5.5: Precision@5 per sector against the random base rate. Retrieval exceeds the base rate in every sector; the gap (lift) is largest for energy and health, whose vocabulary is most distinctive, and smallest for the broad consumer sector.

Why some sectors retrieve better: a qualitative look. The per-sector gap has a concrete explanation, visible in individual queries. In technology, different companies genuinely share themes: a headline about NVIDIA's data-centre accelerators retrieves, from the Microsoft, Meta and Alphabet feeds, articles about the same wave of AI-chip competition (Case Study 3), so cross-ticker precision is high. The consumer sector behaves differently. Looking at the raw neighbourhood (before the cross-ticker constraint is applied), a query about Walmart's grocery guidance retrieves mostly Walmart's own items plus a macro "US retail sales" story carried on bank feeds: adjacent in theme, but not a consumer analogue from another consumer company. Once the query's own ticker is excluded, as the metric requires, little genuinely relevant material is left, which is why the cross-ticker lift for consumer is low. A query about Coca-Cola retrieves almost only other Coca-Cola items, about its pricing power and beverage volumes, which simply do not resemble Walmart's store-traffic news. The shared label "consumer" is too broad: its members' news is idiosyncratic, so cross-ticker analogues are scarce and the lift over the base rate is small. This is the pattern the numbers show, and it points to a clear refinement for future work: grouping by finer-grained themes rather than by sector.

5.4 Case Study 3: End-to-End Alert and Explanation Faithfulness

To illustrate the news trigger end to end, consider a new headline for NVIDIA, *"Nvidia raises guidance on surging demand for AI data-centre accelerators"*. The system embeds it, retrieves the most similar precedents from the knowledge base built from the real news corpus, and renders the alert below (top five precedents, one-day horizon):

```
News alert for NVDA
"Nvidia raises guidance on surging demand for AI data-centre
accelerators"
```

```
Potential impact (from 5 similar past events): average 1-day move:
−1.97%
Historical precedents:
- 2026-06-24 MSFT (sim 0.68) +1d −3.46%: "Qualcomm debuts line of AI
data center chips..."
- 2026-06-24 NVDA (sim 0.68) +1d −1.64%: "Qualcomm debuts line of AI
data center chips..."
- 2026-06-24 META (sim 0.68) +1d −2.65%: "Qualcomm debuts line of AI
data center chips..."
- 2026-06-24 GOOGL (sim 0.64) +1d −0.46%: "This AI Infrastructure
Stock Could Be Bigger Than Nvidia..."
- 2026-06-24 NVDA (sim 0.58) +1d −1.64%: "Tecan Accelerates
Data-Driven Lab Journey... NVIDIA"
Note: precedents are retrieved by semantic similarity; the impact is
the observed past outcome, not a price prediction.
```

Every retrieved precedent concerns AI data-centre chips and competition for NVIDIA, even though the articles were filed under different company feeds (Microsoft, Meta, Alphabet and NVIDIA itself). Unlike the retrieval metric of Section 5.3, the live alert does not exclude the query's own ticker: that exclusion exists only to keep the evaluation non-trivial, whereas a real user is well served by the most similar precedents, whichever company they concern. Strikingly, the same story (a rival entering the AI data-centre chip market) surfaces across three different company feeds, which the engine correctly groups as one theme. This is direct evidence that retrieval is driven by *meaning* rather than the company name or exact keywords, the very behaviour the correlation engine is designed for. In this instance the precedents happen to be uniformly negative (a competitive-threat cluster), and their average is reported as an observed past outcome, not a prediction. The alert ends with an explicit disclaimer, keeping the system within its no-forecasting scope.

What the retrieval does and does not capture. This example also exposes an honest limitation. The query headline reads as positive (guidance raised), yet the precedents it retrieves form a competitive-threat cluster whose one-day outcomes are uniformly negative. There is no contradiction once the mechanism is clear: sentence-embedding similarity matches on *theme*, here "AI data-centre chips", and not on sentiment or expected direction. The mean impact the alert reports is therefore evidence about how a theme has tended to move, not a directional forecast for this particular item, which is exactly why the alert always lists the individual precedents and closes with a no-prediction disclaimer rather than leaning on a single averaged number. Treating that average as a prediction would be the over-reliance that the trust literature warns against (Bansal et al. 2021; J. D. Lee and See 2004), and the design guards against it by showing the evidence rather than issuing a verdict.

Two cosmetic artefacts of the shallow recent corpus are also visible and worth naming plainly: the precedents share a single retrieval date, because the free news tier returns only a recent window and there is little genuine history to draw on, and the same ticker can appear twice with an identical impact, because same-ticker precedents aligned to the same event day necessarily share their forward return. Both follow from corpus depth, not from the method, and both disappear once the full multi-year knowledge base is built. For the same reason, the positive mean impact obtained for a similar Nvidia query on the bundled sample base in Section 3.3.2 (+6.5% at five days) and the negative mean here (−1.97% at one day) are not in tension: they use different knowledge bases, horizons and encoders, and neither is a forecast.

Explanation faithfulness. Faithfulness is guaranteed by construction: the alert text is rendered directly from the same objects the system computes (the z-score, window and threshold for the anomaly path, and the exact retrieved precedents with their similarity scores and measured impacts for the news path), so the explanation cannot diverge from the underlying computation. This is checked programmatically by an automated test that asserts the rendered alert reproduces exactly each retrieved precedent's date, ticker and similarity score and introduces none that were not retrieved; it is the kind of transparent, design-time explainability advocated in the XAI literature (Barredo Arrieta et al. 2020). Usefulness to a retail investor is inherently subjective; a small rubric-based human assessment (clarity, completeness, actionability) is planned but has not yet been conducted, and is therefore reported as a limitation rather than a result.

5.5 Discussion and Threats to Validity

The three case studies support the core design: volatility-normalised detection is consistent across the market where a fixed threshold is not, semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines, and the end-to-end alert is faithful to its own computation. All results rest on real data and are reproducible from versioned scripts.

The results also have clear limits, which are set out below under the four standard categories of validity so that nothing is glossed over.

Construct validity: do the metrics measure what is claimed? Two proxies carry weight here. Retrieval relevance is judged by a *sector* label, an automatic stand-in for the human judgement of "analogous", so a same-sector precedent on an unrelated topic counts as a hit; the qualitative inspection in Section 5.3 shows this proxy is reasonable but imperfect. Likewise, the anomaly "extreme move" label is volatility-relative, so it is not an independent ground truth, which is why the primary anomaly argument is the label-free *firing-rate consistency*, not agreement with that label. Finally, the event-study figure measures the price change that *followed* a news item, which is an outcome, not a causal effect of the news; it is reported as evidence, never as a forecast.

Internal validity: could something else explain the effect? The retrieval advantage might, in principle, be inflated by trivially matching a company's news to its own past news; this is removed by construction, because the evaluation forbids same-ticker precedents and measures *cross-ticker* precision only. Lookahead is excluded throughout: every quantity uses information available at the time, and impacts are accumulated strictly forward from the event-day close. The main residual threat is confounding (a price move in a post-event window may be driven by market-wide factors rather than the news), which the short windows (+1, +3, +5 days) limit but do not eliminate, and which the no-causation framing acknowledges openly.

External validity: how far do the findings generalise? The study covers fifteen large-capitalisation US tickers across five sectors over a recent window, so the numbers need not transfer to small-capitalisation stocks, to other markets, or to other periods, and news coverage is thinner outside large caps. The corpus is a recent window from a free service rather than the multi-year FNSPID history; the full corpus is identified as the natural way to broaden this scope.

Statistical-conclusion validity: are the comparisons sound? The retrieval results are averaged over five random samples with standard deviations reported (the error bars in Figure 5.4), and the gap between semantic retrieval and every baseline is large relative to that spread; still, five runs on one corpus is a modest sample, and no formal significance test is claimed. The anomaly comparison is reported descriptively, as a difference in firing rates across instruments, rather than as a hypothesis test.

One limit is deliberate rather than a weakness: the system makes no price prediction and evaluates no trading profit, so those questions are out of scope by design. The full FNSPID corpus, a human relevance and usefulness study, and a multi-year impact analysis are the natural next steps, and are carried forward to the conclusions.

5.6 Chapter Conclusions

Across three case studies on real data, CLARION's anomaly trigger fired far more consistently than a fixed threshold, its correlation engine retrieved sector-analogous precedents well above every baseline and independently of the embedding model, and its explanations were faithful to the underlying computation by construction. The next chapter draws these findings together against the research questions.

Chapter 6

Conclusions

This dissertation set out to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers and delivered through Telegram. This chapter summarises what was achieved against the research questions, revisits the contributions, and states the limitations and the directions they open.

6.1 Main Conclusions

The work is best summarised by returning to the three research questions of Chapter 1.

RQ1: transparent detection of abrupt movements. The answer is yes. A rolling z-score detector flags abnormal returns and, more importantly, does so at a steady rate *across stocks of very different volatility*, where a fixed-percentage threshold cannot. As reported in Chapter 5, the firing rate varied by only 0.015 across the fifteen tickers for the z-score versus 0.344 for the fixed baseline, and the z-score reached an F_1 of 0.516 against an extreme-move proxy, more than double the baseline. The method is simple enough that every alert can be explained to the investor.

RQ2: analogous precedents without lookahead. The answer is affirmative for retrieval. Semantic retrieval with Sentence-BERT recovered markedly more sector-analogous precedents than every trivial alternative: a cross-ticker precision@5 of 0.514 ± 0.015 (over five runs), against 0.346 for a lexical baseline, 0.240 for random and 0.126 for recency. A per-sector breakdown showed the lift is greatest where domain vocabulary is most distinctive (energy, health). The end-to-end case study showed it surfacing genuinely topical precedents across different company feeds. Impact is measured strictly after the event, in the style of an event study, so no future information leaks into the evidence. The measurement machinery is in place and behaves coherently; a large-scale, multi-year validation of impact magnitudes remains for future work.

RQ3: faithful and useful explanations. Faithfulness is achieved: because each alert is rendered directly from the same objects the system computes (the z-score and its parameters, or the retrieved precedents with their scores and measured impacts), the explanation cannot diverge from the underlying logic, and every alert closes with an explicit no-prediction disclaimer. Usefulness to a retail investor is plausible by design but has not yet been measured with a human study, and is therefore reported as an open point rather than a settled result.

6.2 Research Questions and Objectives Achieved

The general objective, to design, implement and evaluate an explainable, reproducible alerting system driven by two triggers, was met: CLARION is a working system whose two triggers were proven end to end and whose two core components were evaluated on real data. The supporting objectives were also met: a thin end-to-end slice was delivered first; each component was kept in its simplest defensible form; the evaluation followed a design fixed in advance; and the whole system is reproducible from versioned scripts. RQ1 and RQ2 are answered affirmatively on real data, while RQ3 is answered for faithfulness and left partly open for usefulness, pending a human study.

6.3 Contributions Revisited

As an Artificial Intelligence *Engineering* dissertation, the contribution is the integration, application and critical evaluation of existing components into a working, explainable and reproducible whole, rather than a new algorithm. Three concrete contributions stand out. First, a documented and reproducible methodology for **news–market impact correlation** that combines sentence-embedding retrieval with event-study impact windows, with explicit choices of similarity metric and horizon and explicit avoidance of lookahead. Second, **CLARION**, an **XAI-first alerting system** whose entire reasoning chain (detection, evidence, sources and precedents) is exposed to a non-expert user and is faithful by construction. Third, a **critical, honest evaluation** of each core component against trivial and lexical baselines on real data, with results, ablations and limitations reported transparently and reproducibly from versioned scripts.

6.4 Limitations

Several limitations remain. The evaluation used a recent, few-month news corpus from a free service rather than the multi-year FNSPID history, which bounds the impact analysis and makes the retrieval figures, though real, preliminary. The relevance proxy is sector agreement, an automatic stand-in for human judgement, and the anomaly proxy label is volatility-relative (hence the label-free firing-rate argument is given primacy). News is represented by short headlines only, which limits the semantics that can be captured. Because embedding similarity captures theme rather than direction, a retrieved cluster can mix positive and negative outcomes, so the mean impact is evidence about a theme and not a directional signal, which is why the individual precedents are always shown alongside it. The retrieval result is averaged over five runs, but no human usefulness study has yet been conducted. Finally, and by design, the system performs no price prediction or trading, so profit-based questions are out of scope rather than unanswered.

6.5 Future Work

The limitations map directly onto future work. The most immediate step is to build the full multi-year knowledge base from FNSPID, for which the streaming pipeline is already in place, enabling a richer impact analysis with dispersion and direction-consistency statistics. A small human study, applying a clarity–completeness–actionability rubric to a set of alerts, would address the open question of usefulness (RQ3). The retrieval could be strengthened by embedding full article bodies rather than headlines and by comparing further embedding models, and the explanation could be optionally enriched with financial sentiment or feature attribution where it adds value. As a product, the system would also rank alerts by severity and rate-limit the least important, de-duplicate near-identical precedents, and flag directional disagreement within a retrieved cluster, directly addressing the alert-fatigue and over-reliance risks discussed in Chapter 4. Together these would turn a sound prototype into a more thoroughly validated system, without giving up the simplicity and transparency that make it defensible in the first place.

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Appendix A

Reproducibility

This appendix records the engineering detail needed to reproduce the system and its results, kept out of the main text so that the body reads as a dissertation rather than a software specification.

A.1 Environment

The system is implemented in Python 3.12, chosen for the maturity of its packages for the machine-learning stack. Dependencies are pinned and frozen in a lock file so the environment can be recreated across machines. The numerical and data handling rely on established open-source libraries; sentence embeddings use the Sentence-BERT framework on a CPU build of the underlying deep-learning library, installed from a dedicated index to avoid the much larger GPU download; figures are produced with a standard plotting library. Table A.1 lists the pinned versions of the core dependencies; the complete set is frozen in the repository's lock file.

Table A.1: Pinned versions of the core dependencies (Python 3.12).

Library	**Version**
numpy	2.1.3
pandas	2.2.3
matplotlib	3.11.0
scikit-learn	1.9.0
yfinance	1.4.1
sentence-transformers	5.6.0
transformers	5.12.1
torch (CPU build)	2.12.1
pytest	8.3.4
ruff	0.8.6

A.2 Repository Organisation

The code is organised as one package per component, mirroring the architecture of Figure 4.1: market data, news retrieval, anomaly detection, the correlation engine (similarity and event study), the historical knowledge base, the explanation engine, and alert delivery, with a small orchestration layer wiring them into the two end-to-end flows. Pure logic is separated from input/output, and heavy or networked dependencies are loaded lazily, so the numerical core is unit-tested without a network connection or the model stack. Secrets are read only from a local, ignored environment file and are never written to versioned files.

A.3 Tests and Reproducible Results

The system is covered by an automated test suite spanning the anomaly detector, the event study, similarity, the knowledge base, the news parser, the explanation engine and the evaluation modules, together with an offline end-to-end smoke test of the news trigger. Tests that require network access or heavy models are marked and excluded from the default run so that it is fast, deterministic and offline. All experimental results in Chapter 5 are regenerated by scripts with a fixed random seed, and the result figures are written directly into the thesis figure directory, closing the loop between code and document. The three case studies are reproduced by three commands, each writing its table and figure into the evaluation and figure directories:

- Precedent retrieval (precision@k, Case Study 2): `python scripts/evaluate.py`
- Per-sector retrieval breakdown (Case Study 2): `python scripts/evaluate_per_sector.py`
- Anomaly firing-rate and window ablation (Case Study 1): `python scripts/evaluate_anomaly.py`

The data-fetch step that precedes them, and the full build of the historical knowledge base, are documented in the repository's operator guide.

A.4 Data and Attribution

Large data and models are never versioned; they are recreated by the data scripts, and only small samples are committed. The FNSPID dataset (Dong, Fan, and Peng 2024) is used under its CC BY-SA 4.0 licence, with attribution as required; the live market-data and news services are used within their free tiers.